

Data Plots: Isokinetic Temperature in Oxidation of Double Bonds

This exploration provides discussion that will be applicable for answering question #23 in chapter 8 of the textbook¹

The Isokinetic Temperature Controversy

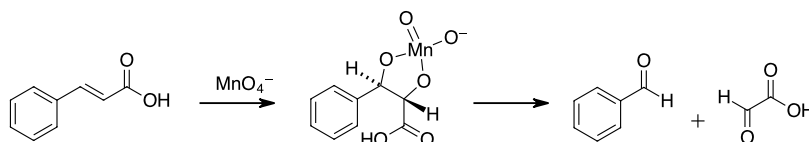
As I was exploring the idea of entropy-enthalpy compensation as described in the textbook,¹ I came across many literature reviews and reports expressing suspicion and casting shade on the idea.² Eyring plots in general are a statistical nightmare and the values of $\Delta S^\ddagger$ and $\Delta H^\ddagger$ so determined are highly correlated due to the math of the plot and not necessarily due to chemical reality.

Despite this, an isokinetic temperature is often observed in experimental Hammett series,³ so it is taken as a real phenomenon by physical organic chemists. I will leave a detailed discussion of the criticism to policy wonks, statisticians and physical chemists.

The first example of a system with an observed isokinetic temperature that I found was from a paper by Perez-Benito⁴. It was published in a forgotten European journal from before the internet. Fortunately, the librarian was able to find it for me via inter-library loan.⁵

The Reaction and the Data

I obtained a set of thermodynamic data for the oxidative cleavage of double bonds in cinnamic acids using permanganate.⁴



The authors report thermodynamic data collected between 0 °C and 25 °C for a series of substituted cinnamic acids. We can use this data to determine the value for the isokinetic temperature. As described in the textbook,¹ the isokinetic temperature is the slope of a plot of $\Delta H^\ddagger$ vs $\Delta S^\ddagger$ shown in figure 1 on the next page.

We can also plot the Eyring plots according to the data in table 1 on page 3. The result is presented in figure 2 on the next page. We can see the apparent convergence to an isokinetic point. Do you believe in the isokinetic temperature?

This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*.

¹ Read chapter 8.6.

² For an introduction to the controversy, see "Enthalpy-Entropy Compensation Effect in Chemical Kinetics and Experimental Errors: A Numerical Simulation Approach." J.F. Perez-Benito, M. Mulero-Raichs, *J. Phys. Chem. A*, **2016**, 120, 7598-7609, [10.1021/acs.jpca.6b08079](https://doi.org/10.1021/acs.jpca.6b08079) and references therein.

³ "The isokinetic relationship." W. Linert, R.F. Jameson, *Chem. Soc. Rev.*, **1989**, 18, 477-505. <https://doi.org/10.1039/CS9891800477>

⁴ "Substituent Effects on the Oxidation of Cinnamic Acid by Methyltributylammonium Permanganate in Methylene Chloride." J.F. Perez-Benito, *Chemica Scripta*, **1987**, 27, 433-436. No DOI or URL available. Obtained via inter-library loan. Ask me for a copy.

Scheme 1: Oxidative cleavage of an alkene by permanganate.

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⁵ Always start a project early. Inter-library loans take two to five days to produce the paper needed. If you encounter the need for the librarian's help, you will also need to have that time available.

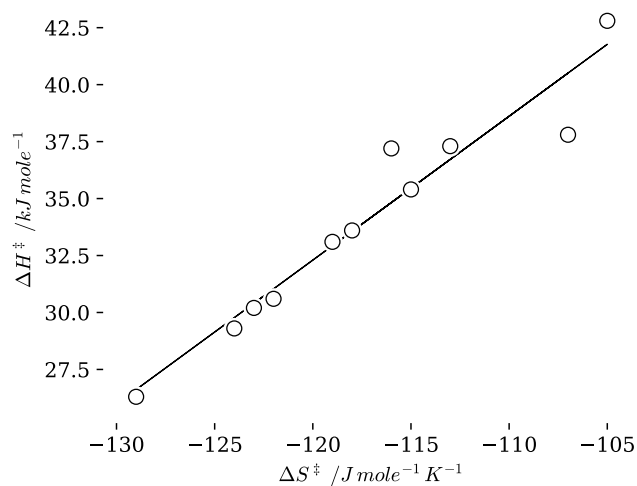


Figure 1: Plot of $\Delta H^\ddagger$ vs $\Delta S^\ddagger$.

The slope is $(631 \pm 56) K$

The two outlier points are for the $\alpha\text{-CH}_3$ and $p\text{-OCH}_3$ reactants. Delete them from the data set and see what happens. Does that increase or decrease your confidence in the idea of isokinetic temperature?

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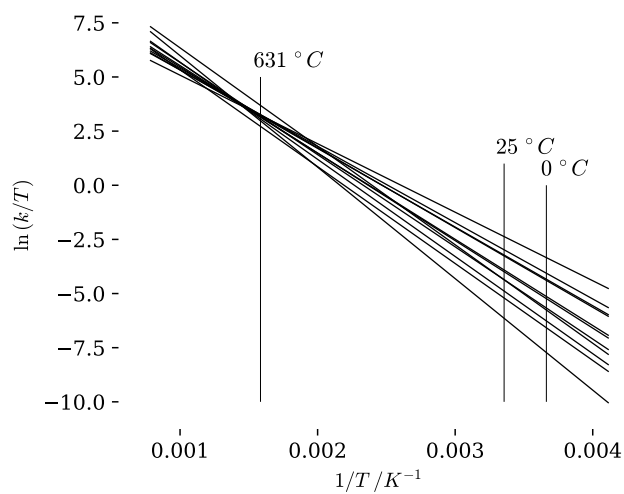


Figure 2: Eyring plot using the thermodynamic data reported in table 1 on the following page. The isokinetic temperature is indicated on the plot along with the experimental temperature range.

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The two plots on this page were generated using the *Python* notebook available at <https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class.17/isokinetic.ipynb>

Substituent	$\Delta H^\ddagger / kJ\ mole^{-1}$	$\Delta S^\ddagger / J\ mole^{-1}\ K^{-1}$
H	33.6 ± 1.1	-118 ± 4
α -D	33.4 ± 0.6	-119 ± 2
β -D	33.1 ± 1.3	-120 ± 5
α -CH ₃	37.2 ± 0.7	-116 ± 3
α -Ph	42.8 ± 1.4	-105 ± 5
<i>o</i> -OCH ₃	37.3 ± 0.9	-113 ± 4
<i>m</i> -OCH ₃	33.1 ± 1.9	-119 ± 7
<i>p</i> -OCH ₃	37.8 ± 2.2	-107 ± 8
<i>m</i> -Cl	30.2 ± 1.2	-123 ± 5
<i>m</i> -Br	30.6 ± 2.1	-122 ± 8
<i>m</i> -NO ₂	26.3 ± 1.9	-129 ± 7
<i>m</i> -CF ₃	29.3 ± 3.6	-124 ± 14
<i>p</i> -CH ₃	35.4 ± 1.4	-115 ± 5

Table 1: Thermodynamic data reported for oxidative cleavage of cinnamic acid by permanganate. Data is from reference 4. Errors are “maximum possible errors” calculated using methodology described in the paper and its references.

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The Math

In figure 1, I plotted $\Delta H^\ddagger$ vs $\Delta S^\ddagger$ and stated that, if a linear relationship is observed, the slope is the isokinetic temperature. In figure 2, I plotted the lines that result from Eyring plots using the thermodynamic parameters reported in the paper and reproduced in table 1. There, I observed a point where most of the rates are equal, at a temperature very close to the one determined in figure 1. If I remove the two outliers, the calculated isokinetic temperature is almost exactly at this nexus point.⁶

In the textbook,⁷ the authors establish the relationship by using the enthalpy-entropy compensation effect.⁸ I will interpret their mathematical argument below.

Let us consider a set of molecules in a Hammett series. Each has a substituent, X. I measured the rates of reaction in a range of temperatures and, by using Eyring plots, obtained values for $\Delta H^\ddagger$ and $\Delta S^\ddagger$ in each case. The differences between the unsubstituted and any substituted case is...

$$\Delta\Delta H^\ddagger = \Delta H_X^\ddagger - \Delta H_0^\ddagger \quad (1)$$

$$\Delta\Delta S^\ddagger = \Delta S_X^\ddagger - \Delta S_0^\ddagger \quad (2)$$

The enthalpy-entropy compensation effect occurs when $\Delta\Delta H^\ddagger \propto \Delta\Delta S^\ddagger$. If we are observing this, we could say that...

$$\Delta\Delta H^\ddagger = \beta \Delta\Delta S^\ddagger \quad (3)$$

...and we can substitute and get...

$$\Delta H_X^\ddagger - \Delta H_0^\ddagger = \beta (\Delta S_X^\ddagger - \Delta S_0^\ddagger) \quad (4)$$

...which can be rearranged to...

⁶ The *Python* notebook is available (see the legend of figure 2) and you can check this statement and my math.

⁷ Chapter 8.6.2 and 8.6.3

⁸ This is also an issue in supramolecular chemistry. The textbook has a whole chapter devoted to the subject. We could explore it in the second half of the course, depending on student interests. The enthalpy-entropy compensation effect in host-guest binding is discussed on page 216 of the textbook (chapter 4.)

$$\Delta H_X^\ddagger = \beta \Delta S_X^\ddagger + (\Delta H_0^\ddagger - \beta \Delta S_0^\ddagger) \quad (5)$$

$$\Delta H_X^\ddagger = \beta \Delta S_X^\ddagger + c \quad (6)$$

This is a linear equation. If we plot $\Delta H^\ddagger$ vs $\Delta S^\ddagger$, we will get a line with a slope of β and a y-intercept of $\Delta H_0^\ddagger - \beta \Delta S_0^\ddagger$.^{9,10}

Exploration Tips

This is a good beginning to an exploration. I could produce a solid 10-page exploration by answering some of the following questions.

Exploration Questions

1. Controversy? I smell a good argument. I could read the referenced paper and follow the trail of references within. Which side am I on? I should do “my own research” before I decide. I could then write about the statistical issues that led to the isokinetic point being criticized.
2. Regardless of the statistical nature of the data, the isokinetic phenomenon is observed in many Hammett systems. I could learn about enthalpy-entropy compensation and write about how that might lead to observing an isokinetic temperature.
3. I could look for more data, plot the isokinetic plots, and compare these reactions, their mechanisms, and the data. Rather than use the reported $\Delta H^\ddagger$ and $\Delta S^\ddagger$ values, I could replot the rate vs temperature data to obtain my own thermodynamic values. Did they agree with the author’s values? I bet some won’t.¹¹ What are your own estimated standard errors?
4. I could plot the Eyring plots using the uncertainties to show confidence intervals on the plots. Is the isokinetic point lost in a haze of uncertainty?¹²
5. What about that “maximum error” reported in table 1? The authors used that method because of the statistical minefield in which they found themselves. Today, we have *Python* tools that can easily handle complex error propagation.¹³
6. Or perhaps I could examine the evidence in the literature for the mechanism of oxidative cleavage of alkenes. I could search for some examples of Hammett plots and kinetic isotope effect experiments for the oxidative cleavage of alkenes. Selected examples from the body of evidence for the Criegee mechanism¹⁴ would make for a good exploration.

An exploration starts with an interesting paper; it does not stay there. That is why it is called an exploration. Wander around the literature and have fun. Teach me something.

⁹ Note that this y-intercept is a collection of constant values ($\Delta H_0^\ddagger$ and $\Delta S_0^\ddagger$ for the unsubstituted case)

¹⁰ You might also have noticed that equation 8.6.4 in the textbook (page 469) is incorrect. The authors should have used equation 3 from this document. But it makes no difference in the end. Plotting the data using equation 3 will give a line with the same slope of β . The intercept will be zero, but we do not care about the value of the intercept in this analysis. There is an errata available for this textbook at <https://uscibooks.aip.org/books/modern-physical-organic-chemistry/>. All textbooks are full of minor errors. However, this unimportant error is not included in the corrections. Should I send the authors a letter, or am I being overly pedantic?

¹¹ Your line fit algorithm may give a slightly different value. Or the reported value might just be wrong.

¹² That sounds like advanced *Python* plotting. Fortunately, I have a complete set of plotting examples available from an annual presentation that I give on plotting with experimental error. This presentation includes examples of Eyring plots that feature confidence intervals. Steal it all from my GitHub pages at <https://blinkletter.github.io/PythonPresentation/start.html>.

¹³ As long as you trust that black box of code that you are using. But this is not a problem because you will provide your code to the reader. Even if ChatGPT wrote all the code for you (it writes half of my own these days) and you have no idea what’s going on inside your computer, you can still explain what you did by publishing the code. Others can thus easily see how you handled error propagation and agree or disagree with your method. Then they will write a long response to the journal editor explaining all your mistakes. That’s a good thing; that’s what science is. Do good work and provide your detailed methods and observations. *Python* notebooks that document your calculations are an important part of this.

¹⁴ “Criegee Intermediates Beyond Ozonolysis: Synthetic and Mechanistic Insights.” Z. Hassan, M. Stahlberger, N. Rosenbaum, S. Bräse, *Angew. Chem. Int. Ed.*, **2021**, 60, 15138–15152. <https://doi.org/10.1002/anie.202014974>